Heejung Choi

Researcher, Asan Institute of Life Sciences, Asan Medical Center
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Interests

Foundation models for healthcare AI; Biomedical Informatics; Clinical Decision Support Systems; Precision Health; Real-World Data; EHR; Biobank; Computational Biology

Education

M.S., Medical Engineering, AMIST, University of Ulsan Sep 2021– Aug 2023

- Advised for Prof. Young-Hak Kim

B.S., Biology & Bioinformatics, Seoul Women's University Mar 2017– Feb 2021

Research Experience

Researcher Department of Information Medicine, Asan Medical Center Mar 2021–Current

- Analyzed tabular and time-series electronic health records
- Developed AI-based clinical decision support systems
- Evaluated model performance using both internal and external validation datasets
- Applied explainable AI (XAI) techniques for model interpretation using SHAP analysis and attention visualization
- Conducted clinician reader studies to evaluate real-world clinical utility
- Deployed open-source LLMs using parameter-efficient fine-tuning (PEFT) techniques

Undergraduate Intern Jan 2020–Sep 2020
Department of Molecular Cell Biology, Sungkyunkwan University

- Advisored for Prof. Dongryeol Ryu
- Performed data visualization using R from NCBI GEO transcriptomic datasets

Honors & Awards

Study Abroad Scholarship (declined) $80,000 2026–2028

- National Institute for International Education (NIIED)

Medical AI Specialist Training Program in EMR — 1st Prize (among 6 teams) 2022

- Korea Health Industry Development Institute (KHIDI)
- Mentored by Prof. Duckyoung Yoon, Yonsei University College of Medicine

- Developed and validated an emergency department disposition prediction model using the MIMIC-IV database

Outstanding Talent Scholarship 2020

Seoul Scholarship Foundation

Publications

First-author

- Choi H, Kang HJ, Ahn I, Gwon H, Kim Y, Seo H, Cho HN, Han J, Kim M, Kee G, Park S. Machine learning models to predict the warfarin discharge dosage using clinical information of inpatients from South Korea. Scientific reports. 2023 Dec 18;13(1):22461.

- Choi H, Kim Y, Kang H, Seo H, Kim M, Han J, Kee G, Park S, Ko S, Jung H, Kim B. Time series forecasting of weight for diuretic dose adjustment using bidirectional long short-term memory. Scientific Reports. 2024 Jul 31;14(1):17723.

Manuscripts Under Review

- Heejung Choi, Yunha Kim, Jiahn Seo, Gaeun Kee, Soyoung Ko, HyoJe Jung, Byeolhee Kim, Boeun Choi, Ha Yeon Kim, Yoobin Park, Jihoon Yoon, Young-Hak Kim, Jungyo Suh, Tae Joon Jun, Large Language Models versus Traditional Machine Learning for Tabular Clinical Prediction. 2026. Journal of the American Medical Informatics Association (Submitted)

Co-author

- Seo J, Kim Y, Choi H, Kim M, Han J, Kee G, Ko S, Jung H, Kim B, Choi B, Park S. Protecting medical privacy with pretrained language models: named entity recognition-based de-identification in Korean unstructured electronic medical records. npj Health Systems. 2025 Sep 4;2(1):32.

- Kim M, Kim Y, Kang HJ, Seo H, Choi H, Han J, Kee G, Ko S, Jung H, Kim B, Choi B. Leveraging BERT for embedding ICD codes from large scale cardiovascular EMR data to understand patient diagnostic patterns. BMC Medical Informatics and Decision Making. 2025 Aug 11;25(1):300.

- Jung H, Kim Y, Seo J, Choi H, Kim M, Han J, Kee G, Ko S, Kim B, Choi B, Kim AR. Clinical assessment of fine-tuned open-source llms in cardiology: From progress notes to discharge summary. Journal of Healthcare Informatics Research. 2025 Dec;9(4):686-702.

- Han J, Kim Y, Kang HJ, Seo J, Choi H, Kim M, Kee G, Park S, Ko S, Jung H, Kim B. Predicting low density lipoprotein cholesterol target attainment using machine learning in patients with coronary artery disease receiving moderate-dose statin therapy. Scientific Reports. 2025 Feb 13;15(1):5346.

- Kim M, Kim Y, Kang HJ, Seo H, Choi H, Han J, Kee G, Park S, Ko S, Jung H, Kim B. Fine-tuning LLMs with medical data: can safety be ensured?. NEJM AI. 2025 Jan 1;2(1):AIcs2400390.

- Cho HN, Ahn I, Gwon H, Kang HJ, Kim Y, Seo H, Choi H, Kim M, Han J, Kee G, Park S. Explainable predictions of a machine learning model to forecast the postoperative length of

stay for severe patients: machine learning model development and evaluation. BMC Medical Informatics and Decision Making. 2024 Nov 20;24(1):350.

- Cho HN, Jun TJ, Kim YH, Kang H, Ahn I, Gwon H, Kim Y, Seo J, Choi H, Kim M, Han J. Task-specific transformer-based language models in health care: scoping review. JMIR Medical Informatics. 2024 Nov 18;12:e49724.
- Kim Y, Kang H, Seo H, Choi H, Kim M, Han J, Kee G, Park S, Ko S, Jung H, Kim B. Development and transfer learning of self-attention model for major adverse cardiovascular events prediction across hospitals. Scientific Reports. 2024 Oct 8;14(1):23443.
- Seo H, Ahn I, Gwon H, Kang H, Kim Y, Choi H, Kim M, Han J, Kee G, Park S, Ko S. Forecasting Hospital Room and Ward Occupancy Using Static and Dynamic Information Concurrently: Retrospective Single-Center Cohort Study. JMIR Medical Informatics. 2024 Mar 21;12:e53400.
- Kee G, Kang HJ, Ahn I, Gwon H, Kim Y, Seo H, Choi H, Cho HN, Kim M, Han J, Park S. Are polypharmacy side effects predicted by public data still valid in real-world data?. Heliyon. 2024 Jan 30;10(2).
- Gwon H, Ahn I, Kim Y, Kang HJ, Seo H, Choi H, Cho HN, Kim M, Han J, Kee G, Park S. LDP-GAN: generative adversarial networks with local differential privacy for patient medical records synthesis. Computers in Biology and Medicine. 2024 Jan 1;168:107738.
- Seo H, Ahn I, Gwon H, Kang HJ, Kim Y, Cho HN, Choi H, Kim M, Han J, Kee G, Park S. Prediction of hospitalization and waiting time within 24 hours of emergency department patients with unstructured text data. Health care management science. 2024 Mar;27(1):114-29.
- Cho HN, Ahn I, Gwon H, Kang HJ, Kim Y, Seo H, Choi H, Kim M, Han J, Kee G, Jun TJ. Heterogeneous graph construction and HinSAGE learning from electronic medical records. Scientific Reports. 2022 Dec 7;12(1):21152.
- Kim Y, Ahn I, Cho HN, Gwon H, Kang HJ, Seo H, Choi H, Kim KP, Jun TJ, Kim YH. RIDAB: Electronic medical record-integrated real world data platform for predicting and summarizing interactions in biomedical research from heterogeneous data resources. Computer Methods and Programs in Biomedicine. 2022 Jun 1;221:106866.
- Ahn I, Gwon H, Kang H, Kim Y, Seo H, Choi H, Cho HN, Kim M, Jun TJ, Kim YH. Machine learning–based hospital discharge prediction for patients with cardiovascular diseases: development and usability study. JMIR Medical Informatics. 2021 Nov 17;9(11):e32662.
- Gwon H, Ahn I, Kim Y, Kang HJ, Seo H, Cho HN, Choi H, Jun TJ, Kim YH. Self–Training With Quantile Errors for Multivariate Missing Data Imputation for Regression Problems in Electronic Medical Records: Algorithm Development Study. JMIR Public Health and Surveillance. 2021 Oct 13;7(10):e30824.